

EECS 16A Designing Information Devices and Systems I

Discussion 3A

Definition - Linear Dependence (I): A set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly dependent if there exist scalars $\alpha_1, \dots, \alpha_n$ such that $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = \vec{0}$ and not all α_i 's are equal to zero.

Definition - Linear Dependence (II): A set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly dependent if there exist scalars $\alpha_1, \dots, \alpha_n$ and an index i such that $\vec{v}_i = \sum_{j \neq i} \alpha_j \vec{v}_j$.¹ In words, a set of vectors is linearly dependent if one of the vectors could be written as a linear combination of the rest of the vectors.

These two definitions are equivalent. If we would like to prove linear dependence with definition (II), we need to first choose a vector $\vec{v}_i$ and show that it is a linear combination of the other vectors. However, with definition (I), we don't need to try to "single out" a vector to get started with the proof. We can blindly write down the equation $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = \vec{0}$ and begin our proof from there. On the other hand, definition (II) gives us a more intuitive way to talk about redundancy. If a vector can be constructed from the rest of the vectors, then this vector does not contribute any information that is not already captured by the other vectors.

Definition (Linear Independence): A set of vectors is linearly independent if it is not linearly dependent. More specifically, from the first definition of linear dependence we can deduce that a set of vectors $\{\vec{v}_1, \dots, \vec{v}_n\}$ is linearly independent if $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = \vec{0}$ implies $\alpha_1 = \dots = \alpha_n = 0$.

For more details, please refer to [Note 3](#).

Properties of Matrix-Vector Multiplication:

- (a) $\mathbf{A}(\vec{x} + \vec{y}) = \mathbf{A}\vec{x} + \mathbf{A}\vec{y}$ (Distributivity)
- (b) $\mathbf{A}\vec{x} \neq \vec{x}\mathbf{A}$
- (c) For $\mathbf{A}\vec{x}$ to be valid, if $A \in \mathbb{R}^{m \times n}$, then $\vec{x}$ must $\in \mathbb{R}^n$.
- (d)

$$\begin{bmatrix} | & | & | & | \\ \vec{a}_1 & \vec{a}_2 & \dots & \vec{a}_n \\ | & | & | & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \vec{a}_1 + x_2 \vec{a}_2 + \dots + x_n \vec{a}_n = \sum_{i=1}^n x_i \vec{a}_i$$

¹In case you are unfamiliar with this notation, the $\sum$ symbol is simply shorthand for addition. For instance, $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n$ can be written as $\sum_{i=1}^n \alpha_i \vec{v}_i$ or $\sum_i \alpha_i \vec{v}_i$, which is a sum over all possible i values. In this instance, $\sum_{j \neq i} \alpha_j \vec{v}_j$ is the sum over all $\alpha_j \vec{v}_j$ excluding the $\alpha_i \vec{v}_i$ term, which can also be calculated as $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n - \alpha_i \vec{v}_i$.

1. Proofs

- (a) Suppose for some non-zero vector $\vec{x}$, $\mathbf{A}\vec{x} = \vec{0}$. Prove that the columns of $\mathbf{A}$ are linearly dependent.
- (b) Suppose there exists a matrix $\mathbf{A}$ whose columns are linearly dependent. Prove that if there exists a solution to $\mathbf{A}\vec{x} = \vec{b}$, then there are infinitely many solutions.
- (c) Suppose there exist two unique vectors $\vec{x}_1$ and $\vec{x}_2$ that both satisfy $\mathbf{A}\vec{x} = \vec{b}$, that is, $\mathbf{A}\vec{x}_1 = \vec{b}$ and $\mathbf{A}\vec{x}_2 = \vec{b}$. Prove that the columns of $\mathbf{A}$ are linearly dependent.

2. Span Proof

Given some set of vectors $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$, show the following:

$$\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} = \text{span}\{\alpha \vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}, \text{ where } \alpha \text{ is a non-zero scalar.}$$

In other words, we can scale our spanning vectors and not change their span.